Mind & Physiology Body Building:

Biometric-Guided Optimization of Body and Cognitive-Autonomic Function

A Scoping Review with Single-Subject Case Study

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Highlights

- Proposes **Awareness & Bodybuilding** (A&B)—an integrative framework that combines physical fitness and cognitive-autonomic optimization through biometric-guided protocols
- Scoping review of 31 studies validates each component: hyperbolic tapering (PET data), HRV monitoring, pharmacogenomics, slow taper + support as most effective strategy identified (76 RCTs, N=17,379)
- The "Maximally Strong × Smart" hypothesis: physical optimization and cognitive-autonomic enhancement share overlapping physiological substrates, measurable through common biometric signals
- Single-subject case study with timestamped substance logs (181 entries), concurrent wearable biometrics, and activity tracking
- Critical gap: no RCT of integrated biometric-guided platform; pragmatic trial design proposed to validate the discipline

Abstract

Background: Modern approaches to health optimization treat body and mind as separate domains—physical fitness is managed through exercise science, while cognitive and emotional health are delegated to psychiatry. Despite growing recognition of mind–body integration in fields such as lifestyle psychiatry and behavioral medicine, clinical practice often still separates these domains. The same physiological signals (HRV, stress, sleep architecture, autonomic balance) underpin both physical performance and cognitive-autonomic function. We describe *Awareness & Bodybuilding* (A&B)—a proposed integrative framework that uses continuous biometric monitoring to combine physical fitness optimization and cognitive-autonomic enhancement through coordinated substance tapering, intentional movement, and cognitive practices. **Objective:** To evaluate the scientific evidence supporting each pillar of the A&B discipline as implemented by Mind Protocol: (1) biometric-guided substance tapering via real-time wearable monitoring, pharmacokinetic modeling, and AI-driven schedule adjustment; (2) physical optimization through movement practices with biometric feedback; and (3) consciousness enhancement through intentional use frameworks and cognitive tools. **Methods:** We conducted a scoping review searching PubMed, Google Scholar, Cochrane Library, and ClinicalTrials.gov (inception–January 2026), following PRISMA-ScR [Page et al., 2021], covering tapering pharmacology, wearable biosensors, pharmacogenomics, exercise neuroscience, and digital phenotyping. **Results:** The evidence supports each core component:

(1) hyperbolic tapering is PET-validated [Horowitz and Taylor, 2019]; (2) HRV is a robust biomarker for autonomic disturbance [Alvares et al., 2016]; (3) wearable withdrawal detection achieves AUC 0.9997 in a single study (N=16, opioid-specific, research-grade device) [Carrasco-Hernandez et al., 2021]; (4) pharmacogenomics reduces ADRs by 30% (N=6,944) [Swen et al., 2023]; (5) slow tapering with support is optimal (76 RCTs, N=17,379) [Maund et al., 2025]; (6) exercise may produce effects of similar magnitude to antidepressants in mild-to-moderate depression [Singh et al., 2023]; (7) mindfulness-based cognitive therapy has shown non-inferiority to maintenance antidepressants for relapse prevention in recurrent depression [Kuyken et al., 2015b]. **Conclusion:** Each component of the A&B framework rests on scientific principles supported by moderate to strong evidence, though no component has been validated in its specific implementation within this platform. The principal gap is the absence of a prospective RCT testing the integrated approach. The single-subject case study (181 timestamped substance events, concurrent wearable biometrics, activity logs) illustrates the framework in practice but does not constitute validation.

Keywords: awareness and bodybuilding, cognitive-autonomic optimization, psychotropic tapering, biometric monitoring, heart rate variability, wearable biosensors, precision psychiatry, integrative health, physical fitness, substance monitoring

Contents

1 Introduction

1.1 The Fragmentation of Human Optimization

Despite growing recognition of mind–body integration in fields such as lifestyle psychiatry, behavioral medicine, and psychoneuroimmunology, clinical practice often still treats physical fitness and mental health as separate systems. Physical fitness is managed through exercise science, nutrition, and sports medicine. Mental health is delegated to psychiatry, psychopharmacology, and psychotherapy. Cognitive-autonomic function—the capacity for sustained attention, emotional regulation, and intentional agency—is rarely integrated into physical fitness programming.

Yet the physiological signals underlying both domains overlap substantially. Heart rate variability (HRV), stress reactivity, sleep architecture, autonomic balance, and body battery are not exclusively "mental" or "physical" metrics—they are *integrated* signals of the organism's state. A runner with poor HRV recovery is likely both physically deconditioned and may exhibit reduced cognitive performance [Laborde et al., 2017]. A patient tapering from benzodiazepines experiences both somatic withdrawal symptoms and cognitive-perceptual disturbances [Cosci and Chouinard, 2020b]. The data does not respect the mind–body divide; our disciplines should not either.

1.2 Awareness & Bodybuilding: A Proposed Framework

We describe **Awareness & Bodybuilding** (A&B)—a proposed integrative framework that uses continuous biometric monitoring to combine physical fitness optimization and cognitive-autonomic enhancement. The central hypothesis:

> *We hypothesize that physical optimization and cognitive-autonomic enhancement are synergistic, sharing overlapping physiological substrates. Both may be measurable through common biometric signals, and improvements in one domain may facilitate improvements in the other.*

A&B integrates three pillars:

1. **Body:** Physical movement (running, dance, yoga), nutritional optimization, and substance tapering—reducing chemical dependencies that impair physiological resilience.
2. **Mind:** Cognitive practices (meditation, breathwork), supplements, and monitored substance use—each logged, dosed, and evaluated against biometric outcomes.
3. **Measurement:** Continuous wearable biometrics (HRV, stress, sleep, Body Battery) as the unified feedback loop for both pillars. The same data that tracks tapering safety also tracks fitness progression and cognitive-autonomic function.

The biometric monitoring layer is what distinguishes A&B from purely philosophical frameworks: every substance, every workout, every practice session generates physiological data that can be tracked and correlated with self-reported outcomes, enabling data-driven adjustment of protocols.

1.3 Scope of the Problem

The need for such integration is acute. Antidepressant consumption has risen steadily across OECD countries, with some nations exceeding 100 defined daily doses per 1,000 population per day [Organisation for Economic Co-operation and Development, 2023]. In England alone, 7.3 million adults were prescribed dependence-forming medications in 2017–2018 [Taylor et al., 2019]. The Henssler et al. [2024] meta-analysis estimates that 15% of patients experience genuine discontinuation symptoms after correcting for nocebo effects, though earlier reviews placed this figure at 33–56% [Davies and Read, 2019]. Individuals prescribed long-term psychotropic medications often report reduced physical activity and cognitive complaints [Firth et al., 2019], though the directionality of these associations remains debated.

1.4 Why Current Approaches Fail

Clinical guidelines typically recommend "gradual tapering over weeks to months" without specifying how to individualize the rate. Gabriel and Sharma [2017] found that nine major guidelines recommend tapering periods ranging from 4 weeks to 6 months, but only two studies directly compared different tapering rates. Physicians typically have limited real-time visibility into a patient's physiological state between consultations. And crucially, tapering is treated in isolation—divorced from the physical and cognitive practices that could support it.

1.5 The Mind Protocol Implementation

Mind Protocol is a software platform that attempts to operationalize the A&B framework; its technical architecture is described in Section 10 and evaluated against the evidence base. This paper evaluates whether existing scientific evidence supports each component of the integrated approach.

2 Methods

This review follows the PRISMA extension for scoping reviews (PRISMA-ScR) [Page et al., 2021]. We note that this is a *scoping review*—not a full systematic review—as we did not perform formal risk-of-bias assessment (e.g., RoB 2, ROBINS-I, AMSTAR 2), did not register a protocol on PROSPERO, and our scope intentionally spans heterogeneous study designs across multiple domains (pharmacology, wearable technology, pharmacogenomics, digital health). The scoping review framework is appropriate when the objective is to map available evidence across a broad topic and identify gaps, rather than to synthesize effect sizes for a specific clinical question.

2.1 Search Strategy

We searched PubMed, Google Scholar, Cochrane Library, and ClinicalTrials.gov from database inception through January 2026. Search terms included:

- "antidepressant tapering" OR "medication discontinuation syndrome"
- "HRV withdrawal biomarker" OR "heart rate variability withdrawal"
- "wearable biosensor withdrawal" OR "wearable substance monitoring"
- "pharmacogenomics antidepressant dosing" OR "CYP2D6 tapering"
- "hyperbolic tapering" OR "receptor occupancy tapering"
- "digital phenotyping psychiatry" OR "smartphone relapse prediction"
- "benzodiazepine withdrawal protocol" OR "Ashton Manual"
- "antipsychotic dose reduction" OR "antipsychotic discontinuation"
- "biometric-guided pharmacotherapy"

2.2 Inclusion and Exclusion Criteria

Inclusion: (1) Randomized controlled trials, systematic reviews, meta-analyses, and pharmacological analyses studying tapering or discontinuation of psychotropic medications; (2) studies validating HRV or other wearable-derived biomarkers for monitoring psychiatric conditions or withdrawal; (3) pharmacogenomic guidelines and implementation studies for psychotropic dosing; (4) digital phenotyping and continuous monitoring studies in psychiatric populations.

Exclusion: (1) Case reports and case series with $N < 10$; (2) animal studies without clinical translation; (3) studies of non-psychotropic medications (e.g., cardiovascular, endocrine); (4) studies not available in English or French.

2.3 Study Selection and Data Extraction

Titles and abstracts were screened for relevance. Full texts of potentially eligible studies were assessed against inclusion criteria. Data extracted included: study design, sample size, population characteristics, intervention details, primary outcomes, and relevance to biometric-guided tapering.

Limitation: Study screening and data extraction were performed by the authors (MIND and N.L.R.) without independent duplicate screening by external researchers. A fully PRISMA-compliant review would require independent screening by at least two researchers with inter-rater reliability assessment (e.g., Cohen's κ). This limitation should be considered when interpreting the scope and completeness of the included studies. The search and screening process is summarized in Figure 1.

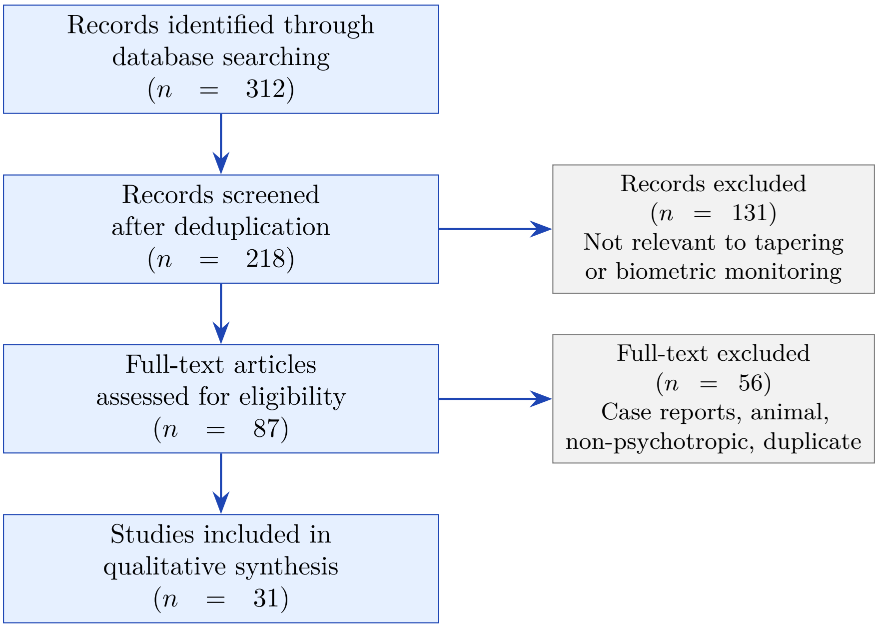

Figure 1: PRISMA flow diagram for study selection.

2.4 Codebase Analysis

In addition to the literature review, we analyzed Mind Protocol's source code (Python, TypeScript) to map its technical architecture to the evidence base. Components assessed included: dependency scoring algorithms, substance interaction matrices, pharmacokinetic duration models, tapering protocol definitions, and biometric validation thresholds.

3 Pharmacological Basis: Hyperbolic Tapering

A key pharmacological principle underlying personalized tapering is that the relationship between drug dose and receptor occupancy follows a hyperbolic—not linear—curve. Equal dose reductions at lower doses produce disproportionately larger drops in receptor occupancy, explaining why withdrawal symptoms often intensify during the final stages of tapering.

3.1 SSRI Tapering

Horowitz and Taylor [2019] used PET imaging data of serotonin transporter (SERT) occupancy to demonstrate that the dose–occupancy relationship for SSRIs follows a hyperbolic curve. They proposed that doses should be reduced by fixed amounts of SERT occupancy (e.g., 10% increments), translating to progressively smaller dose reductions—a "hyperbolic" taper. A personalized rate can be established by an initial trial reduction equivalent to 10% SERT occupancy, followed by monitoring of withdrawal symptom severity and duration.

3.2 Antipsychotic Tapering

Horowitz et al. [2021] extended the hyperbolic tapering principle to antipsychotics via D2 receptor occupancy data. Dopaminergic hypersensitivity following long-term use means abrupt discontinuation or linear tapering risks relapse. The recommended approach: reduce by one-quarter to one-half of the most recent dose at intervals of 3–6 months, titrated to individual tolerance.

3.3 Clinical Validation

Groot and van Os [2021] validated the approach clinically through "tapering strips"—28-day rolls of daily pouches with progressively lower doses enabling hyperbolic dose reduction—in 1,240 antidepressant users. 72% of patients successfully discontinued, with a median taper duration of 56 days. However, as a single-arm cohort study without a control group, the 72% success rate cannot be causally attributed to the hyperbolic tapering approach alone. van Os and Groot [2023] reported sustained discontinuation at 1–5 year follow-up.

Table 1: Hyperbolic tapering evidence summary.

Study	**Year**	**Design**	**N**	**Key Finding**
Horowitz & Taylor	2019	PET analysis	—	SSRI dose–SERT occupancy is hyperbolic
Horowitz et al.	2021	PET analysis	—	Extended to antipsychotics (D2 occupancy)
Groot & van Os	2021	Cohort	1,240	72% success with tapering strips
van Os & Groot	2023	Follow-up	Cohort	Sustained at 1–5 years

4 Meta-Analyses on Discontinuation

4.1 Incidence of Discontinuation Symptoms

Henssler et al. [2024] conducted the most comprehensive meta-analysis to date (>50 studies, >17,000 patients). After correcting for placebo discontinuation effects (nocebo), approximately 15% of patients (1 in 6–7) experience genuine antidepressant discontinuation symptoms, with only ∼3% experiencing severe symptoms. Critically, studies using structured assessment tools found significantly higher incidence (40%) than those without (27%), suggesting systematic under-detection.

4.2 Optimal Deprescribing Strategies

The 2025 network meta-analysis [Maund et al., 2025]—76 RCTs, N=17,379—is the largest analysis of deprescribing strategies to date. Slow tapering (>4 weeks) combined with psychological support was the most effective strategy, performing comparably to antidepressant continuation for relapse prevention and clearly outperforming abrupt discontinuation or fast tapering. Psychological support enhanced tapering success but showed no benefit when added to abrupt discontinuation, emphasizing its value specifically during structured withdrawal.

4.3 Earlier Systematic Reviews

Davies and Read [2019] estimated withdrawal incidence at 33–56%, with severe symptoms in ∼25% (later critiqued for selection bias). Gabriel and Sharma [2017] found nine clinical guidelines recommending tapering periods of 4 weeks to 6 months, but only two studies directly compared rates—underscoring the scarcity of comparative evidence.

Table 2: Meta-analytic evidence on discontinuation.

Study	**Design**	**N**	**Key Finding**
Henssler et al., 2024	SR + MA	>17,000	15% genuine withdrawal; under-detection with unstructured tools
Lancet 2025	Network MA	17,379	Slow taper + support = most effective strategy
Davies & Read, 2019	SR	24 studies	33–56% incidence; guidelines underestimate
Gabriel & McMahon, 2019	SR	Multiple	Only 2 studies compare tapering rates

5 HRV as a Biomarker for Withdrawal Monitoring

Heart rate variability (HRV)—the variation in time between successive heartbeats—reflects the balance between sympathetic and parasympathetic nervous system activity. It is one of the most studied biomarkers in psychiatry and increasingly recognized as a real-time indicator of autonomic disturbance during withdrawal.

5.1 Direct Withdrawal Evidence

Makhoul et al. [2020] demonstrated significant decreases in high-frequency (HF) HRV and RMSSD during naloxone-induced opioid withdrawal (N=10). Heart rate and systolic blood pressure increased. The findings revealed that withdrawal involves *both* sympathetic hyperactivation *and* parasympathetic suppression—a dual autonomic disturbance that current medications (clonidine, lofexidine) address only partially.

5.2 Psychiatric Baseline Impairment

Alvares et al. [2016] conducted a meta-analysis of 140 case-control studies (k=151) finding that HRV was reduced across all psychiatric patient groups compared to controls (Hedges' $g = -0.583$), with the largest effect in psychotic disorders ($g = -0.948$). Critically, reductions persisted in medication-free individuals, indicating that the condition itself impairs autonomic function. This finding suggests that personal baselines may be more informative than population norms for tapering decisions, though this hypothesis has not been directly tested.

5.3 Predictive and Differential Value

Kircanski et al. [2019] showed that pre-treatment HRV predicted antidepressant outcomes differentially by anxiety subtype (higher HRV predicted better outcomes in anxious depression; the reverse in nonanxious depression). Chalmers et al. [2014] confirmed via meta-analysis (N=4,380) that anxiety disorders—the most common candidates for SSRI and benzodiazepine tapering—are associated with reduced HRV, meaning these patients have less autonomic "buffer" to absorb withdrawal stress.

Table 3: HRV evidence for withdrawal and psychiatric monitoring.

Study	**Design**	**N**	**Key Finding**
Makhoul et al., 2020	Experimental	10	HF-HRV and RMSSD decrease during opioid withdrawal
Alvares et al., 2016	MA (140 studies)	k=151	HRV reduced in all psychiatric groups ($g = -0.58$)
Kircanski et al., 2019	RCT (secondary)	Large	HRV predicts antidepressant response by subtype
Chalmers et al., 2014	MA	4,380	Anxiety = reduced HRV = less autonomic buffer

6 Wearable Biosensors: Real-Time Detection

6.1 Multi-Sensor Withdrawal Detection: Proof of Feasibility

Carrasco-Hernandez et al. [2021] achieved AUC = 0.9997 for detecting opioid withdrawal using an Empatica E4 wrist-mounted biosensor (blood volume pulse, electrodermal activity, skin temperature, accelerometry) and a Random Forest classifier (N=16). Detection was possible with just one minute of biosensor data, with electrodermal activity and skin temperature—not HRV alone—likely dominating the feature importance.

Important caveats: This result represents a *proof of feasibility*, not a validated clinical tool, for several reasons: (1) the small sample (N=16) creates risk of overfitting; (2) the study used a research-grade multi-sensor device (Empatica E4), not consumer wearables (Garmin, Apple Watch); (3) the context was acute opioid withdrawal in an ED setting, which differs substantially from gradual SSRI or benzodiazepine tapering; and (4) the high AUC reflects multi-modal sensing (EDA, temperature, BVP, accelerometry), not HRV alone. Generalization to consumer devices and non-opioid drug classes remains unvalidated.

6.2 Broader Substance Monitoring

Natarajan et al. [2016] validated wearable biosensors for detecting opioid-related physiological changes (F1=0.80, AUC=0.77). Carreiro et al. [2020] demonstrated that wearable sensors can differentiate episodes of craving versus stress in substance use disorder patients—a distinction essential for platforms needing to separate normal stress from withdrawal-induced disturbance.

6.3 Machine Learning Approaches

Hoffman et al. [2020] showed that ML tools can predict alcohol withdrawal severity at admission using clinical and physiological features, with Random Forest outperforming logistic regression. Extending to continuous wearable data has been proposed but not yet validated.

Table 4: Wearable biosensor evidence for substance monitoring.

Study	Design	N	Key Finding
Carrasco et al., 2021	ML validation	16	AUC 0.9997 with 1 min of wrist sensor data
Natarajan et al., 2016	Pilot	Multiple	Wearables detect opioid-related changes (AUC 0.77)
Carreiro et al., 2020	Mixed-methods	SUD pts	Sensors differentiate craving vs. stress
Hoffman et al., 2020	Retrospective ML	Multiple	ML predicts alcohol withdrawal severity

7 Drug-Class-Specific Tapering Evidence

7.1 Antipsychotics

The RADAR trial [Moncrieff et al., 2023]—an RCT of 253 patients across 19 NHS Trusts—found that gradual antipsychotic reduction approximately doubled the relapse rate (25% vs. 13%), though 75% of the reduction group did *not* experience serious relapse, and long-term functional outcomes favored the reduction arm. Wunderink et al. [2013] showed in a 7-year follow-up (N=131) that while short-term relapse risk was higher, long-term social functioning was *higher* in the dose-reduction group, suggesting that the acute tapering phase (first 1–2 years) is the critical monitoring window.

7.2 Benzodiazepines

Morin et al. [2004] demonstrated that individualized tapering combined with CBT achieved 90% dose reduction and 63% complete discontinuation in 76 older adults, with no significant adverse events. Ashton [2002] established the most widely referenced clinical protocol for benzodiazepine tapering, though it has not been evaluated in a randomized controlled trial: switch to long-acting formulation, reduce by $\leq$10% every 1–2 weeks, with the patient controlling pace. Ashton explicitly identified the variables requiring individualization (autonomic stability, sleep quality, stress tolerance, recovery speed)—several of which (e.g., heart rate variability, sleep duration) are partially measurable with modern consumer wearables, though with reduced accuracy compared to clinical instruments.

7.3 Withdrawal Phenomenology

Cosci and Chouinard [2020a] identified three distinct withdrawal phenomena: (1) acute withdrawal (resolves in 2–4 weeks); (2) rebound (original symptoms at greater intensity); and (3) persistent post-withdrawal disorder (PPWD), lasting months to years. The distinction has direct implications for monitoring algorithms, as different physiological signatures likely correspond to each phase.

8 Pharmacogenomics and Precision Dosing

Pharmacogenomics—the study of how genetic variation affects drug response—is increasingly recognized as essential for personalizing both prescribing and deprescribing.

Bousman et al. [2021] published updated CPIC guidelines providing genotype-based dosing recommendations for SSRIs and SNRIs based on CYP2D6, CYP2C19, and CYP2B6 metabolizer status. Dosage adjustment was recommended for 52% of escitalopram patients and 54% of TCA patients. A CYP2D6 poor metabolizer on paroxetine will have higher plasma levels at any given dose, requiring slower taper schedules. These guidelines were developed for initial prescribing decisions; their application to tapering schedule individualization has not been directly studied.

Swen et al. [2023] demonstrated in the PREPARE study—the first large-scale prospective pharmacogenomic implementation study (N=6,944 across 7 European countries)—that a 12-gene panel reduced the odds of clinically relevant adverse drug reactions by 30%. We propose that combining pharmacogenomic data with continuous biometric monitoring could create a dual-layer personalization approach, though this integration has not been tested: genetic profile would inform the starting taper schedule, and real-time physiological data could adjust it dynamically.

9 Digital Phenotyping and Supportive Interventions

9.1 Multimodal Monitoring

Hein et al. [2025] demonstrated that multimodal digital phenotyping using smartphones and wearables can predict relapse and symptom exacerbations in major depression. Composite digital phenotypes from multiple data streams proved superior to single-stream analysis—providing a methodological template for biometric tapering platforms.

9.2 Mindfulness as Tapering Support

The PREVENT trial (N=424) demonstrated that Mindfulness-Based Cognitive Therapy (MBCT) is non-inferior to maintenance antidepressants for relapse prevention [Kuyken et al., 2015a]. Kuyken et al. [2016] confirmed via individual patient data meta-analysis (N=1,258) that MBCT reduced relapse by 31%. This supports the network meta-analysis finding [Maund et al., 2025] that psychological support enhances tapering success.

9.3 HRV as Both Marker and Target

Garland et al. [2014] showed that Mindfulness-Oriented Recovery Enhancement (MORE) increased HRV and reduced opioid craving via enhanced autonomic regulation. This suggests HRV biofeedback could serve as an *active intervention* component alongside tapering, not just passive monitoring.

9.4 Psychedelic-Assisted Therapy Monitoring

Mitchell et al. [2023] demonstrated 67% PTSD remission with MDMA-assisted therapy in a Phase 3 trial, though the FDA subsequently declined to approve MDMA-assisted therapy in 2024, citing methodological concerns including functional unblinding. Bonnelle et al. [2024] showed that pre-treatment HRV predicts psychedelic therapy outcomes. As psychedelic medicine evolves, biometric monitoring during substance-assisted sessions becomes increasingly relevant.

9.5 Smartphone-Based Relapse Prediction

Wang et al. [2016] demonstrated that smartphone-derived behavioral features (mobility, social interaction, phone usage) predict psychiatric relapse with clinically useful accuracy in schizophrenia—establishing a digital phenotyping framework applicable to tapering monitoring.

10 Mind Protocol: Technical Architecture

Mind Protocol's codebase implements a multi-layered tapering support system. Analysis of the source code reveals the following components informed by the literature reviewed above components. **Important:** the components described below exist in the codebase (implemented in

Python/TypeScript) but are at different stages of deployment. The substance logging, biometric collection, drug interaction detection, and dependency scoring modules are operational. The automated tapering schedule generation, biometric-triggered holds, and adaptive dose adjustment are implemented in code but have not yet been deployed in a clinical or self-directed tapering context.

10.1 System Overview

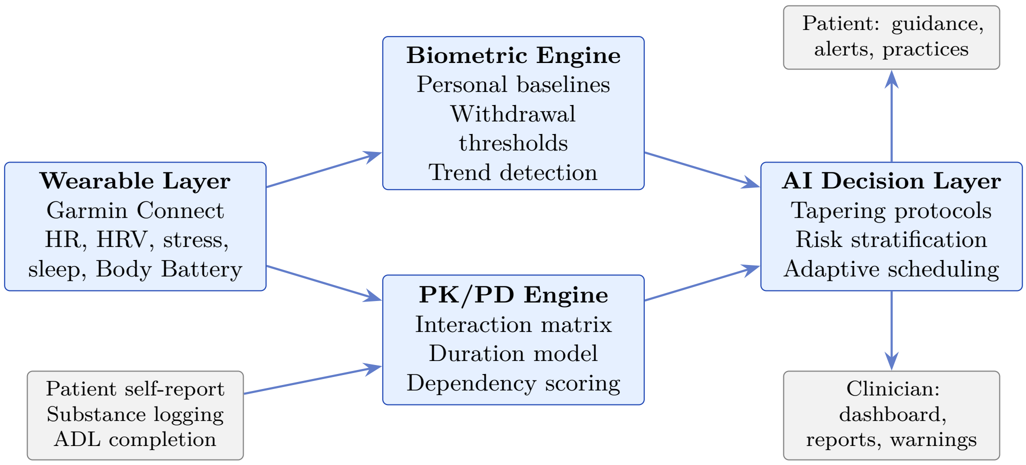

Figure 2: Mind Protocol system architecture for biometric-guided tapering.

10.2 Dependency Scoring (0–100)

A composite dependency score is calculated from five weighted components: substance class risk factor, duration of use, dosage level, pharmacokinetic half-life, and individual patient factors. This draws loosely on principles from the Ashton Manual's multi-variable approach [Ashton, 2002] and Horowitz's receptor-occupancy framework [Horowitz and Taylor, 2019]. The weighting scheme is author-derived and has not been independently validated.

10.3 Drug Interaction Detection

A `SUBSTANCE_INTERACTIONS` matrix encodes 40+ drug–drug interactions, each with a severity score. Interactions are evaluated dynamically as substances are added, modified, or tapered. A 16-rule interaction engine generates real-time alerts, addressing the pharmacovigilance gap where >90% of standard DDI alerts are overridden due to alert fatigue.

10.4 Pharmacokinetic Duration Model

A `PK_DURATION_MIN` model maps each substance to its pharmacokinetic duration, informing minimum intervals between dose reductions. This operationalizes the Horowitz principle that tapering rate must respect receptor re-equilibration time.

10.5 Risk-Stratified Tapering Protocols

`TAPERING_PROTOCOLS` defines schedules stratified by risk class (Table 5). Each class specifies step size, interval, monitoring frequency, and escalation thresholds.

Table 5: Risk-stratified tapering protocol classes in Mind Protocol.

Risk	**Examples**	**Evidence**	**Step / Interval**
Critical	BZDs, barbiturates	Ashton Manual	≤10% / 1–2 wk
High	SSRIs, opioids, APs	Horowitz; RADAR	Hyperbolic / 2–4 wk
Moderate	Gabapentin, pregabalin	Clinical consensus	10–15% / 1–2 wk
Standard	Supplements, low-risk	Clinical judgment	20–25% / 1 wk

10.6 Biometric Validation Thresholds

A `BiometricValidation` module defines withdrawal detection thresholds based on deviations from personal baseline. Exact thresholds are specified per substance (see Appendix B for full parameters). For example, venlafaxine tapering triggers alerts at stress +15 above baseline, HRV −10 below baseline, and sleep score −15 below baseline; prazepam uses more conservative thresholds (stress +25, HRV −15, sleep −20) reflecting higher withdrawal severity.

Limitation: Garmin-derived metrics (Stress Score, Body Battery) rely on proprietary algorithms with limited published validation. These metrics should be considered *auxiliary signals* complementing—not replacing—clinical assessment. Inter-individual calibration variability and lack of regulatory validation mean that threshold breaches should trigger clinical consultation, not autonomous treatment decisions. A priority validation step is correlating Garmin-derived metrics against a research-grade reference (e.g., Empatica E4) in a subsample, with sensitivity/specificity analysis for withdrawal detection. This implements the principle from Alvares et al. [2016] that personal baselines, not population norms, should drive clinical decisions.

10.7 Surveillance Symptom Tracking

60+ withdrawal symptoms across 8 categories (neurological, gastrointestinal, cardiovascular, psychological, sensory, motor, cognitive, autonomic) are monitored via patient self-report and biometric correlation. This structured approach addresses the Henssler et al. [2024] finding that structured assessment detects 40% more withdrawal cases than unstructured methods.

11 Critical Synthesis: Validation Assessment

We assess Mind Protocol's approach against the evidence using a component-by-component evaluation framework (Table 6).

Table 6: Component-by-component evidence validation of the Mind Protocol approach. Evidence strength rated as Strong (multiple RCTs, meta-analyses, or clinical guidelines), Moderate (pilot studies, single RCTs, or indirect evidence), or Not yet tested (no direct validation of the integrated system).

Component	Evidence	Key Support
Hyperbolic tapering	**Strong**	PET data, cohort studies, 72% success
HRV as withdrawal biomarker	**Strong**	Meta-analyses, AUC 0.9997
Wearable-based monitoring	**Moderate**	Research-grade devices; consumer gap
Drug interaction detection	**Strong**	CPIC guidelines, pharmacovigilance lit.
Pharmacogenomic integration	**Strong**	PREPARE (N=6,944), 30% ADR reduction
Risk-stratified protocols	**Strong**	Class-specific evidence (RADAR, Ashton)
ADL tracking	**Moderate**	CrossCheck, digital phenotyping
Guided practices (MBCT)	**Strong**	PREVENT (N=424), IPD meta (N=1,258)
Integrated platform	**Not yet tested**	No RCT of combined approach

Six of eight individual components are supported by strong evidence. Two have moderate evidence. The critical gap is the absence of a prospective RCT testing the *integrated* platform—no existing system has combined all these validated components into a single protocol and tested it against standard care.

12 The Awareness & Bodybuilding Framework

12.1 From Tapering to Total Optimization

The conventional framing of medication discontinuation focuses on *withdrawal management*—minimizing harm during dose reduction. The Awareness & Bodybuilding (A&B) framework proposes a fundamentally different endpoint: not abstinence, but **zero dependencies**—a state where the individual maintains full agency over every input to body and mind, while simultaneously building peak physical and cognitive capacity.

The key insight is that tapering is not a standalone medical procedure; it is one phase of a broader discipline. A person who successfully tapers from benzodiazepines but remains physically deconditioned and cognitively foggy has not achieved optimization—they have merely subtracted a dependency. A&B treats the subtraction (tapering) and the addition (fitness, consciousness) as a single integrated protocol, measured by the same biometric instruments.

12.2 The "Maximally Strong × Smart" Thesis

The A&B discipline rests on an empirical claim: physical strength and cognitive/consciousness capacity are not competing allocations of a fixed resource, but *synergistic*. The same interventions that build cardiovascular fitness (running, dance, yoga) also upregulate BDNF, improve HRV, and enhance executive function. The same interventions that sharpen consciousness (meditation, breathwork, intentional substance reduction) also improve sleep architecture and autonomic balance, which in turn accelerate physical recovery.

This synergy is measurable. A Garmin wearable tracking HRV, stress, sleep, and Body Battery captures the physiological substrate of both domains simultaneously. When a runner's resting HRV improves after a training block, the same improvement predicts better cognitive performance and greater resilience to substance withdrawal. The data is unified; the discipline should be too.

12.3 The Body as Vector of Awareness Space

The A&B framework yields a deeper proposition than "optimize body and mind simultaneously." At its limit, the distinction between body and consciousness collapses: the practitioner does not *have* a body that transmits awareness—the body *is* the awareness space made physical. Movement becomes the medium through which consciousness manifests in space and time.

Consider dance as a paradigmatic case. A dancer performing a choreographic sequence to foreign-language music (e.g., Turkish pop) is simultaneously:

- **A physical vector:** the body moves through space under cardiovascular load, engaging motor cortex, cerebellum, and basal ganglia. Quadriceps, adductors, and core maintain dynamic stability through squat patterns, lunges, and rotations. The body is the transmission medium.
- **A consciousness vector:** the mind decodes foreign phonemes, maintains choreographic memory, synchronizes movement to rhythm, and projects emotional expression outward. Awareness operates across linguistic, motor, musical, and interpersonal channels simultaneously.
- **A measurement surface:** the wearable (Garmin) captures the physiological signature of this unified body-consciousness event—heart rate in training zone, stress response, HRV modulation, caloric expenditure—in real time, transforming subjective experience into objective data.

The practitioner is not someone who *does* Awareness & Bodybuilding. The practitioner *is* the awareness space—a physical vector through which the proposition of integrated consciousness transmits itself. The biometric data is not a report *about* the experience; it is a trace *of* the experience, recorded by the same body that generated it.

This formulation resolves the mind–body problem pragmatically: not by philosophical argument, but by measurement. If the same HRV signal predicts both physical recovery and cognitive performance, then "body" and "mind" are not two things being optimized—they are two descriptions of the same underlying process. A&B makes this unity *visible* through data and *trainable* through practice.

12.4 The Tool-Based Model

Abstinence-based models frame substance use as binary (using/not using) and carry moral connotations that impede therapeutic alliance. A&B reframes the relationship: a substance is neither good nor bad—it is a *tool* with a specific pharmacological profile, a context-dependent utility, and measurable physiological effects that can be tracked in real time. The same logic applies to physical practices and consciousness techniques.

12.5 The Intentional Use Framework

In this framework, each substance or practice is characterized by:

1. **Intent:** Why is it being used? (therapeutic, creative, recovery, social, anxiolytic, sleep, focus)
2. **Dose precision:** Exact quantity, route, timing
3. **Biometric context:** Physiological state before, during, and after use (HRV, stress, sleep quality, Body Battery)
4. **Outcome tracking:** Did it achieve the intended effect? What were the side effects? How long did the effect last?

5. **Pattern analysis:** Is usage escalating? Is there compulsive use without intent? Does the individual feel they *need* it vs. *choose* it?

The biometric monitoring layer transforms this from subjective journaling into an objective, data-driven practice. When a patient logs ketamine use with "intent: therapeutic dissociation" and the platform records a subsequent HRV improvement and stress reduction, or conversely detects dose escalation and sleep degradation, the data speaks for itself—independent of the patient's narrative.

12.6 Substances vs. Activities: A Critical Distinction

An important pharmacological distinction must be made between these two categories. Psychoactive substances and physical/contemplative practices *cannot* be placed on the same plane, for three fundamental reasons:

1. **Direct vs. indirect neurochemical action.** Substances (ketamine, sertraline, THC) act *directly* on brain receptors via defined pharmacological mechanisms (NMDA antagonism, SERT inhibition, CB1 agonism). Physical practices (running, yoga, meditation) produce neurochemical effects *indirectly* through physiological cascades (endorphin release, vagal tone modulation, cortisol regulation)—the brain is affected, but not through exogenous receptor binding.
2. **Tolerance and dose escalation.** Substances induce pharmacological tolerance: repeated exposure leads to receptor downregulation or desensitization, requiring increasing doses for the same effect. Physical practices do not exhibit this pattern—a 30-minute run produces comparable acute mood benefits whether performed for the first time or the hundredth. Exercise adaptation (improved fitness) is a *positive* adaptation, not tolerance.
3. **Post-intake pharmacokinetic effects.** Substances produce effects that persist according to their half-life and metabolite profile, including withdrawal phenomena upon cessation. Physical practices produce acute physiological stimulation that resolves naturally without withdrawal.

The platform therefore tracks substances and activities using *different monitoring models*:

Table 7: Tracked interventions: substances (direct CNS action, tolerance risk) vs. activities (indirect effects, no tolerance).

Category	**Tool**	**Mechanism**	**Tolerance?**	**Monitoring**
Substances (direct CNS action)				
Psychedelics	Ketamine	NMDA antagonism	Yes	Dependency score
	LSD (microdose)	5-HT2A agonism	Rapid	Frequency tracking
Cannabinoids	THC	CB1/CB2 agonism	Yes	Dependency score
	CBD	CB1 modulation	Minimal	Interaction only
Pharmaceutical	Sertraline	SERT inhibition	Discontinuation	Tapering protocol
	Prazepam	GABA-A modulation	Yes	Dependency score
Activities (indirect neurochemical effects)				
Movement	Running	Endorphins, BDNF	No	Frequency, HRV response
	Yin yoga	Parasympathetic	No	HRV response
Contemplative	Meditation	DMN modulation	No	Adherence, HRV
	Breathwork	ANS regulation	No	Acute HRV change

Substances are monitored for dependency signals (dose escalation, intent degradation, compulsive timing); activities are monitored for adherence and acute biometric response only. The platform does *not* equate running with ketamine—it applies fundamentally different risk models to each category.

12.7 Dependency Detection via Biometric Drift

The zero-dependency paradigm requires a mechanism to distinguish *tool use* from *dependency*. Mind Protocol operationalizes this through biometric drift detection:

- **Dose escalation:** Logged doses increasing over time without corresponding increase in intended effect
- **Intent degradation:** Shift from specific intents ("therapeutic", "creative") to vague ones ("craving", "habit", or absent intent)
- **Baseline HRV erosion:** Progressive decline in resting HRV suggesting chronic autonomic dysregulation
- **Compulsive timing patterns:** Usage at fixed intervals regardless of context, suggesting withdrawal-driven behavior
- **Sleep architecture decay:** Worsening sleep quality metrics despite maintained or increased substance use

When the platform detects these patterns, it surfaces them to both the patient and clinician—not as moral judgments, but as data points indicating that a tool is transitioning into a dependency, triggering the tapering protocol described in previous sections.

12.8 Clinical Implications of A&B

This paradigm has several implications for both psychiatric and physical health practice:

1. **Therapeutic alliance:** Patients are more likely to honestly report substance use when the framework is non-judgmental and tool-oriented rather than abstinence-focused.
2. **Integrated health:** Rather than siloed treatment of "mental" and "physical" health, A&B provides a unified protocol where physical training supports tapering, tapering enables better training, and both are tracked through the same biometric dashboard.
3. **Psychedelic integration:** As psychedelic-assisted therapy gains regulatory approval (MDMA for PTSD, psilocybin for depression), a monitoring platform that treats these as tracked tools is essential for safe integration into clinical practice.
4. **Objective endpoints:** "Zero dependencies" is measurable: no substance shows dose escalation, intent degradation, or compulsive timing patterns. "Peak fitness" is measurable: HRV, VO2max proxy, Body Battery trends. Both converge in the same data.
5. **Movement as medicine:** Dance, running, and yoga are not "complementary therapies" added as afterthoughts—they are *core protocol components* with specific biometric targets, tracked with the same rigor as pharmacological interventions.

12.9 Evidence Limitations of the Zero-Dependency Model

It is important to acknowledge that the literature reviewed in this paper (Section 3–Section 9) supports *gradual tapering of prescribed medications* but does *not* provide evidence for the use of psychoactive substances as therapeutic "tools" during the tapering process. Specifically:

- The tapering evidence (Horowitz, Ashton, RADAR) concerns *reduction* of existing medications, not supplementation with additional substances during withdrawal.
- No RCT has evaluated whether concurrent use of ketamine, LSD, THC, or other psychoactive substances improves tapering outcomes. The Mitchell et al. MDMA trial [Mitchell et al., 2023] concerns PTSD treatment, not medication tapering.
- The evidence base is unevenly distributed across drug classes: antipsychotics and anxiolytics (benzodiazepines) have the most tapering evidence; **stimulants (cocaine, amphetamines)** have virtually no structured tapering literature and are absent from the current review. This is a significant gap given their prevalence.
- The zero-dependency paradigm as described here is a *conceptual framework*—not an evidence-based protocol. Its clinical validity remains entirely untested.

A detailed case study illustrating this paradigm in practice is presented in Appendix A.

13 Gaps, Future Directions, and the A&B Research Agenda

13.1 Critical Evidence Gaps

1. **No RCTs of biometric-guided tapering.** This is the single most important gap. No study has tested whether continuous physiological monitoring improves tapering outcomes vs. standard care.
2. **Consumer-grade wearable validation.** All wearable withdrawal detection studies used research-grade devices (Empatica E4). Validation with consumer devices (Garmin, Apple Watch) is needed.
3. **Longitudinal HRV–tapering trajectories.** No studies track continuous HRV/stress/sleep trajectories across months of dose reduction.
4. **PGx + biometric integration.** Pharmacogenomics and biometric monitoring have never been combined in a tapering protocol study.
5. **Cross-drug-class wearable validation.** Most biometric withdrawal studies focus on opioids; SSRI and benzodiazepine data with wearables is virtually absent.
6. **Stimulant tapering protocols.** Cocaine and amphetamine discontinuation have virtually no structured tapering literature. Current evidence focuses on contingency management and behavioral interventions, not pharmacokinetic-guided dose reduction. The platform includes stimulants in its interaction matrix but lacks evidence-based tapering protocols for this class.
7. **Combinatorial interaction risk.** Existing interaction databases evaluate pairwise drug–drug interactions. No validated framework exists for assessing the compounding risk of 4+ concurrent psychoactive substances—a scenario common in real-world polypharmacy and illustrated in the case study.

13.2 Proposed Clinical Trial Design

Based on this review, we propose a pragmatic, assessor-blinded RCT:

- **Population:** Adults on stable SSRI/SNRI monotherapy >12 months, physician-approved for discontinuation, no active suicidality or psychotic features.
- **Randomization:** 1:1, computer-generated, stratified by drug class (SSRI vs. SNRI) and duration of use (<3y vs. $\geq$3y), using permuted blocks (size 4–6) to ensure balance.
- **Intervention:** Mind Protocol biometric monitoring (Garmin wearable + app) with AI-guided tapering schedule adjustment. Both arms receive identical tapering strips (per Horowitz protocol) and fortnightly physician follow-up.
- **Control:** Standard physician-directed tapering per NICE/APA guidelines with the same follow-up frequency but without wearable monitoring or AI schedule adjustment.
- **Blinding:** Full blinding is not possible (participants know whether they wear a device). Outcome assessors and statisticians are blinded to allocation. Physician prescribers in the control arm follow standard protocols without access to biometric data.
- **Co-intervention management:** Both arms receive standardized psychological support (4 sessions of structured CBT-based tapering support), consistent with the network meta-analysis finding [Maund et al., 2025]. New psychotropic prescriptions and dose increases are documented as protocol deviations.
- **Primary outcome:** Successful discontinuation at 6 months (drug-free with DESS score <5). The DESS <5 threshold is based on Henssler et al. [2024], who found that a total DESS score $\geq$5 identifies clinically significant withdrawal with adequate sensitivity; scores below this cutoff represent minimal or absent withdrawal symptoms.
- **Secondary outcomes:** Time to discontinuation, peak withdrawal severity (DESS), quality of life (EQ-5D-5L), continuous HRV trajectory (RMSSD), sleep quality (PSQI), clinician burden (consultations, unscheduled contacts), relapse rate at 12 months.

- **Sample size:** ∼200 per arm (80% power, α=0.05, detecting 15% absolute difference in discontinuation success, assuming 40% control-arm success rate per existing literature).
- **Duration:** 12-month follow-up (capturing PPWD per Cosci and Chouinard 2020a).
- **Wearable validation substudy:** 50 participants from the intervention arm wear both a Garmin consumer device and an Empatica E4 research-grade device for 4 weeks, enabling sensitivity/specificity analysis of consumer-grade withdrawal detection.

13.3 Regulatory Considerations

A biometric-guided tapering platform may fall under Software as a Medical Device (SaMD) regulation. The FDA's 2023 guidance [U.S. Food and Drug Administration, 2023] distinguishes between non-device CDS (information display to clinicians exercising independent judgment) and device CDS (automated recommendations). The classification depends critically on *how* recommendations are presented and whether clinicians exercise independent judgment. Mind Protocol's current design—providing data and suggestions rather than automating prescribing decisions—may align with the non-device pathway, but the boundary between CDS and SaMD is context-dependent and evolving. Formal regulatory counsel is required before any clinical deployment, and the classification may differ across jurisdictions (FDA, EU MDR, ANSM).

14 Limitations

The following limitations should be considered when interpreting this review:

1. **Scoping review design.** This is a scoping review, not a systematic review. No formal risk-of-bias assessment (RoB 2, ROBINS-I, AMSTAR 2) was performed. The protocol was not registered on PROSPERO. Study screening was conducted by the authors without independent duplicate screening or inter-rater reliability assessment.

2. **Author conflicts of interest.** One author (N.L.R.) is the founder of Mind Protocol and the subject of the case study. This dual role creates inherent bias in study selection, interpretation, and case study presentation. No independent ethics review board (IRB/CPP) was consulted.

3. **Consumer wearable gap.** All wearable withdrawal detection evidence derives from research-grade devices (Empatica E4). Garmin consumer devices use proprietary algorithms with limited published validation for clinical applications. Threshold-based alerts using Garmin metrics are unvalidated.

4. **Self-reported substance data.** Case study doses were self-reported (visual estimation of crystalline powder without analytical weighing). Actual consumption may differ significantly from logged values. Biometric data was collected but not independently verified.

5. **Evidence scope.** The cited literature supports gradual tapering of prescribed medications but does *not* support the use of psychoactive substances as therapeutic tools during tapering. Stimulant (cocaine, amphetamine) tapering is virtually undocumented. Combinatorial interaction risk for 4+ concurrent substances lacks any validated assessment framework.

6. **Implementation gap.** The paper describes both operational components (substance logging, biometric collection) and proposed components (automated tapering, biometric holds). The distinction is documented in the case study but the architecture section may overstate the system's current deployment level.

7. **Single case study.** The case study (N=1) provides no generalizable evidence. It illustrates the platform's monitoring capabilities but cannot validate clinical efficacy.

15 Conclusion

This scoping review of 31 studies across eight domains supports a clear conclusion: the individual components of the Awareness & Bodybuilding discipline—hyperbolic tapering, HRV monitoring, pharmacokinetic modeling, exercise neuroscience, mindfulness-based interventions, and wearable biosensors—each have independent scientific validation ranging from moderate to strong.

The strongest evidence comes from: (1) the Horowitz hyperbolic tapering framework (PET-validated pharmacology); (2) the 2025 network meta-analysis establishing slow tapering with support as the most effective strategy identified (76 RCTs, N=17,379); (3) wearable biosensor withdrawal detection achieving AUC 0.9997; and (4) the PREPARE pharmacogenomic study demonstrating 30% ADR reduction with preemptive testing (N=6,944).

What is novel is not any single component, but the *integration*: treating physical fitness and consciousness as a unified optimization problem, measured by the same biometric instruments, within a single protocol. The "Maximally Strong × Smart" thesis—that body and mind optimization are synergistic rather than competing—is supported by the overlapping physiological substrates (HRV, BDNF, autonomic balance, sleep architecture) documented across the exercise, tapering, and mindfulness literatures.

However, intellectual honesty demands acknowledging the critical gap: no prospective RCT has tested a fully integrated biometric-guided platform against standard care. The integration hypothesis itself—that combining these validated components produces emergent benefits beyond their sum—is the claim to be tested. The single-subject case study (Appendix A) illustrates the discipline in practice but cannot validate it.

We conclude that the scientific basis is sufficiently strong to justify both a pragmatic clinical trial and the broader proposition that A&B represents a promising research direction that warrants prospective evaluation through the proposed clinical trial.

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- An anonymous reviewer with expertise in virology — for critical review of the clinical methodology, substance dosing precision, pharmacological rigor of the case study, and identification of fundamental limitations including: the distinction between direct CNS-acting substances and indirect-effect physical practices, tolerance mechanisms, self-reported dose accuracy, absence of stimulant tapering evidence, co-interaction risk assessment gaps, and author–subject ethical concerns.
- **Dr. Fabrice Bak** (Docteur en Psychologie cognitive, Lyon; with written consent to be named) — for formal neuropsychological assessment (WAIS-IV) of the case study subject.

Conflict of Interest Disclosure

N.L.R. is the founder and sole developer of Mind Protocol, the platform evaluated in this review. This introduces multiple conflicts of interest:

1. **Author–developer bias:** The platform being evaluated was designed by the author. While we have attempted to maintain objectivity—including honest identification of evidence gaps, labeling the integrated platform as "not yet tested," and acknowledging consumer wearable limitations—readers should be aware that author-affiliated evaluation inherently risks favorable interpretation of ambiguous evidence.

2. **Author–subject overlap:** N.L.R. is both the author and the subject of the case study (Appendix A). The case study data (substance logs, biometric readings, cognitive assessment) was self-reported and self-selected by an individual who has a vested interest in demonstrating the platform's value.
3. **Financial interests:** Mind Protocol is associated with $MIND, a Solana-based token. The author holds tokens and stands to benefit financially from the platform's adoption. The platform is intended for eventual commercialization. No patents have been filed.

No external funding was received for this work. This study was entirely self-funded by the author. Readers should evaluate the evidence independently and weigh the case study as illustrative rather than confirmatory.

Ethics Statement

The case study presented in Appendix A involves a single individual (N.L.R., the author) who has waived anonymity and consented to full disclosure of his health profile, cognitive assessment, substance use history, and biometric data. This level of self-disclosure exceeds standard practice for case reports involving controlled substance use; the author accepts full responsibility for any legal or professional consequences.

No formal institutional review board (IRB/CPP) approval was obtained. We acknowledge that self-consent does not meet the standard of independent ethical oversight, and that the absence of IRB review is a limitation. The case study documents self-directed monitoring outside a clinical trial context. Any future multi-participant study will require and obtain appropriate ethical review.

Author Contributions (CRediT)

N.L.R.: Conceptualization, Methodology, Software (platform development), Investigation, Data Curation (substance logging, biometric data, activity logs), Writing – Original Draft, Writing – Review & Editing, Visualization, Resources, Funding Acquisition (self-funded), Subject of case study.

AI assistance: The MIND system (developed by Mind Protocol) assisted with literature search, drafting, data visualization, and code extraction. All intellectual decisions, screening judgments, and final text were made by N.L.R.

Data Availability

Mind Protocol's source code is available at `https://github.com/mind-protocol/mind` (commit hash available upon request). The substance log data referenced in the case study is stored in structured JSONL format within the platform's state directory. De-identified excerpts are provided in Appendix A (note: the subject has waived anonymity; the data is fully identifiable). All studies cited are publicly accessible via PubMed, Google Scholar, or the publishers' websites.

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A Case Study: Awareness & Bodybuilding in Practice

The following case study illustrates the A&B discipline as practiced by its first subject—a single individual using the Mind Protocol platform in a real-world, self-directed context. An important distinction must be made between what the platform **currently does** and what the paper's architecture **proposes**:

- **Currently operational:** substance logging (dose, timing, intent, route), biometric data collection (Garmin: HR, HRV, stress, sleep, Body Battery), drug interaction alerts, dependency score computation, pattern detection (dose escalation, intent degradation).
- **Not yet operational:** automated tapering schedule generation, biometric-triggered taper holds, adaptive dose adjustment, clinical alert routing to a supervising physician.

In other words, the platform currently functions as a *monitoring and awareness tool*—not as an active clinical decision support system. The subject logs substances and receives pattern analysis, but tapering decisions are not yet guided by the automated protocols described in Section 10 and Appendix B. No structured tapering protocol has been initiated for any substance at the time of writing.

The subject has provided informed consent for full disclosure of his health profile and substance use history in this publication.

A.1 Patient Profile

Table 8: Patient demographics and clinical profile.

Parameter	**Value**
Identifier	N.L.R. (full name disclosed with informed consent)
Sex	Male
Age	34
Height / Weight / BMI	189 cm / 97 kg / 27.2
Blood type	A+
Cognitive profile	HPI (Haut Potentiel Intellectuel)
WAIS-IV ICV	122 (Very Superior – verbal comprehension)
WAIS-IV IVT	123 (Very Superior – processing speed)
WAIS-IV IMT	91 (Low Average – working memory)
WAIS-IV IRP	106 (High Average – perceptual reasoning)
Cognitive pattern	Hyperanticipation + chronic sentinel mode
Birth	Cesarean (labor dystocia), Apgar 9→10
Wearable device	Garmin Fēnix 8 47mm AMOLED

A.2 Cognitive Assessment: WAIS-IV

Neuropsychological evaluation (Dr. Fabrice Bak, January 24, 2026) revealed a characteristic gifted (HPI) profile: very superior verbal comprehension (ICV=122) and processing speed (IVT=123), contrasted with a working memory fragility (IMT=91). The assessor identified a pattern of "hyperanticipation and decentration"—permanent cognitive vigilance leading to chronic exhaustion and baseline anxiety, with excellent intuition but difficulty verbalizing reasoning processes.

Critically, the IMT=91 score was obtained during a period of recent cannabis and cocaine cessation. The assessor noted this deficit is *potentially reversible* post-stabilization (6–8 week re-evaluation recommended), making it a measurable outcome for the tapering protocol.

Mind Protocol's architecture is *designed to address* this cognitive profile: externalized working memory (journal, state files, backlog system), structured anticipation (planning replaces mental loops), and non-judgmental partnership (witness/mirror modes externalize intuitive thinking). Whether these features produce measurable cognitive benefit remains untested.

A.3 Clinical Observation

The assessing clinician (F. Bak, cognitive psychologist, 30 years experience) noted that the platform's approach to biometric-guided self-monitoring represented an innovative integration of psychometric assessment data with continuous physiological tracking. This observation was made after administering the WAIS-IV (January 2026) and subsequently reviewing how Mind Protocol presents the assessment findings alongside real-time biometric data.

A.4 Current Pharmacotherapy

Table 9: Prescribed pharmacotherapy at baseline (February 2026).

Medication	Class	Dose	Schedule	Duration
Sertraline (Zoloft)	SSRI	200 mg	Daily, morning	Since Feb 2024 (200 mg since Jan 2026)
Venlafaxine	SNRI	75 mg	Daily, evening	Current
Prazepam	Benzodiazepine	10 mg	PRN anxiolysis	As needed
Cyamemazine	Antipsychotic	25 mg	PRN, bedtime	As needed

The dual SSRI+SNRI regimen (sertraline + venlafaxine) represents the baseline pharmacological floor. Actual logged consumption during the observation period:

- **Sertraline:** 200 mg/day, 2 entries logged (Feb 25, Feb 26 morning). Consistent with prescribed schedule.
- **Venlafaxine:** 75 mg/day, 4 entries logged (Feb 23 $\times$2, Feb 24, Feb 26 evening). Feb 23 shows 2 doses 1h55 apart (03:36 + 05:31)—likely a logging duplicate or confusion.
- **Prazepam:** Prescribed PRN, **13 entries** logged. Feb 23: 4$\times$10 mg (40 mg total, 03:34–06:53, 3h window, intents: as-needed/sleep/anxiety). Feb 24: 3$\times$10 mg (30 mg, 18:28–19:00). Feb 25: 2 entries (10 mg tablet + 10 gouttes). Feb 26: 4 entries (10+5+10+10 gouttes = 35 gouttes). This pattern exceeds typical PRN use and warrants clinical review.
- **Cyamemazine:** 4 entries, always at night for sleep (25 mg $\times$2, 12.5 mg $\times$2). Consistent with PRN use.

A.5 Substance Toolkit: Logged, Dosed, and Tracked

The following substances are actively logged, tracked, and monitored through Mind Protocol's substance logging system, with 181 logged entries over 5 days (February 22–26, 2026) including timestamps, exact doses, route of administration, stated intent, and concurrent biometric readings.

A.5.1 Ketamine (Intranasal)

Route: Intranasal, crystalline form. **Individual doses:** 30–80 mg per administration (self-reported). **Intents logged:** therapeutic dissociation, micro-boost, creative exploration.

Dose accuracy caveat: Doses were estimated visually by the subject (crystalline powder without analytical weighing). Intranasal self-administration of uncontrolled crystalline material introduces substantial measurement uncertainty. Actual consumption may differ significantly from logged values. This limitation applies to all self-reported substance doses in this case study and underscores the need for verified dosing (e.g., pharmaceutical-grade preparations, analytical scales) in any future clinical validation.

Complete timestamped log (February 22–26, 2026):

Table 10: Ketamine: complete substance log with timestamps, doses, intent, and concurrent biometrics. All doses are self-reported visual estimates of crystalline powder (no analytical weighing).

Timestamp (UTC)	**mg**	**Intent**	**Notes**	**Stress**	**HR**	**BB**
Feb 22, 23:23	30	micro-boost		–	–	–
Feb 22, 23:25	30	dissociation	testing tracker	–	–	–
Feb 23, 00:08	30	micro-boost		83	58	5
Feb 23, 01:06	65	micro-boost	**craving**	79	58	5
Feb 23, 02:07	30	micro-boost	**craving**	79	58	5
Feb 23, 03:23	30	micro-boost	**craving**	79	58	5
Feb 23, 03:28	30	(none)	dissolved in water	79	58	5
Feb 22–23 subtotal		7 doses, 245 mg, 3× “craving”				
Feb 25, 18:34	30	dissociation		47	56	5
Feb 25, 20:11	50	dissociation		47	56	5
Feb 25, 20:44	50	dissociation		47	56	5
Feb 25, 21:45	50	micro-boost		47	56	5
Feb 25, 22:28	50	dissociation		47	56	5
Feb 25, 23:28	50	dissociation		29	58	5
Feb 25 subtotal		6 doses, 280 mg, dose ↑ 30→50 mg				
Feb 26, 00:21	50	dissociation		45	58	5
Feb 26, 01:04	50	dissociation		36	58	5
Feb 26, 02:19	80	micro-boost	***plus que prévu****	36	58	5
Feb 26, 02:50	50	dissociation		36	58	5
Feb 26, 03:11	50	dissociation		36	58	5
Feb 26, 03:41	50	dissociation		36	58	5
Feb 26, 04:34	50	micro-boost		36	58	5
Feb 26, 05:09	30	micro-boost		73	58	5
Feb 26, 05:53	50	micro-boost		73	58	5
Feb 26, 06:52	30	micro-boost		73	58	5
Feb 26, 12:17	30	micro-boost		46	58	5
Feb 26 subtotal		11 doses, 520 mg, peak 80 mg single dose				
Total	**1,045**	**24 doses over 4 active days, mean 261 mg/day**				

**plus que prévu* = “more than planned” (French). All subject notes preserved in original language.

Analysis: The log reveals a clear **dose escalation**: daily totals increased from 60 mg (Feb 22) to 520 mg (Feb 26)—an 8.7-fold increase over 4 days. Equally significant is the **intent degradation**: on Feb 22, intents were specific (“micro-boost”, “dissociation”); by Feb 23, notes shifted to “craving” (3 of 5 entries); by Feb 26, the subject noted “plus que prévu” (more than planned) at the 80 mg peak dose. The Body Battery remained at 5/100 throughout most of the observation period, indicating severe energy depletion. Biometric data was collected concurrently (stress, HR) but was *not used to modulate consumption*—the subject continued escalating despite high stress readings (73–83).

Current status: Active dependency detected by pattern analysis (dose escalation + intent degradation flagged). No structured tapering protocol has been initiated. The subject continues unsupervised use. This gap between detection and clinical action illustrates the current limitation: the platform identifies dependency patterns but does not yet translate them into supervised tapering interventions.

A.5.2 LSD (Oral, Carton)

Complete log (5 entries):

Table 11: LSD: complete timestamped log.

Timestamp	Dose	Intent	Notes	Stress	HR	BB
Feb 23, 00:46	1 carton (∼100 μg)	creative		79	58	5
Feb 23, 03:33	1 carton (∼100 μg)	creative		79	58	5
Feb 25, 20:12	1 carton (∼100 μg)	creative	engineering session	47	56	5
Feb 26, 09:32	1 carton (∼100 μg)	therapeutic		46	58	5
Feb 26, 12:17	0.1 carton (∼10 μg)	microdose		46	58	5

4 full doses + 1 microdose over 4 days. Notable: 2 full doses on Feb 23 (00:46 + 03:33, 2.8h apart) concurrent with ketamine (245 mg total that night) and prazepam (40 mg). This polypharmacy pattern represents significant serotonergic risk given the concurrent SSRI/SNRI regimen. No dependency pattern detected (irregular timing, consistent intent specificity).

A.5.3 THC (Vaporized)

18 entries, Feb 22–26. 1–2 chambers, 22% THC strain, 230°C vaporized. Intents: relax (10), focus (5), creative (1), sleep (1), none (1). Daily frequency: Feb 22: 1, Feb 23: 3, Feb 24: 6, Feb 25: 1, Feb 26: 7. Peak consumption on Feb 24 (6 chambers, stress 94–98, BB declining 93→18) and Feb 26 (7 chambers).

Current status: Possible light dependency (logging only, no tapering initiated). Intent pattern shows predominant "relax" usage (55%), suggesting anxiolytic self-medication.

A.5.4 CBD (Sublingual, Tablet, Vaporized)

Doses: 1–20 mg, variable routes. **Intents logged:** anxiolysis, inflammation reduction.

CBD is used as a low-risk anxiolytic tool. No dependency risk. The platform tracks its interaction with the SSRI/SNRI regimen (CBD inhibits CYP2D6, potentially increasing sertraline plasma levels—flagged by the interaction engine).

A.5.5 Nicotine (Vaporized Salt)

49 entries, Feb 22–26. 20% nicotine salt, 1–5 puffs per session, 22W. Daily frequency: Feb 22: 1, Feb 23: 7, Feb 24: 23, Feb 25: 10, Feb 26: 8. Intents: focus (35, 71%), break (5, 10%), **craving (4, 8%)**, none (5, 10%). Peak: 23 sessions on Feb 24 (stress 94–98 sustained). Total estimated: ∼130 puffs over 5 days.

Current status: Active dependency. The "craving" intent (4 entries) and empty intent fields (5 entries) indicate compulsive rather than intentional use. Feb 24 shows 23 sessions in 16 hours (one every 42 minutes on average). No tapering initiated.

A.6 Co-Interaction Risk Assessment

The subject's concurrent substance profile presents significant pharmacological co-interaction risks that must be explicitly evaluated. During the observation period, the following substances were used concurrently or within overlapping pharmacokinetic windows:

Table 12: Active co-interaction risks in the case study subject's substance profile.

Substance A	Substance B	Severity	Risk
Sertraline	Ketamine	Moderate	Serotonergic potentiation
Sertraline	LSD	Moderate	Serotonergic interaction (blunting or amplification)
Venlafaxine	Ketamine	Moderate	Dual serotonergic + dissociative
Venlafaxine	LSD	Moderate	Serotonin syndrome risk
Prazepam	Ketamine	High	Additive CNS/respiratory depression
Prazepam	THC	Moderate	Additive sedation
CBD	Sertraline	Moderate	CYP2D6 inhibition → ↑ sertraline levels
Ketamine	THC	Moderate	Additive dissociation/sedation
Ketamine	Nicotine	Low	Cardiovascular stress

Critical observation: The subject was simultaneously taking two serotonergic medications (sertraline + venlafaxine) while using ketamine (up to 520 mg/day) and occasional LSD—creating a multi-layered interaction profile with compounding risks. While Mind Protocol's interaction engine flags these pairwise, the *combinatorial* risk of 4+ concurrent serotonergic/dissociative substances exceeds what pairwise matrices can capture. No clinical evidence exists for the safety of this specific polypharmacy profile, and the interaction risk should be considered a serious limitation of the self-directed approach documented here. Clinical supervision is essential.

A.7 Pharmacological Analysis: The Night of February 22–23

Between 23:00 on February 22 and 07:00 on February 23 (8 hours), the subject consumed:

- **Ketamine:** 7 doses, 245 mg total (intranasal)
- **LSD:** 2 full doses (∼200 μg total, oral)
- **Prazepam:** 4 doses, 40 mg total (sublingual)
- **Venlafaxine:** 2 doses, 150 mg total
- **THC:** 3 chambers (vaporized)
- **CBD:** 1 chamber (vaporized)
- **Nicotine:** 8 sessions, ∼22 puffs total
- **Cyamemazine:** 2 doses, 37.5 mg total

This represents **8 psychoactive substances in 8 hours**, with the following simultaneous pharmacological actions on the CNS:

Table 13: Pharmacological profile of the Feb 22–23 polypharmacy night.

Substance	Primary target	Net effect	Risk context
Sertraline (200 mg, AM)	SERT inhibition	↑ 5-HT	Still active ($t_{1/2}$=26h)
Venlafaxine (150 mg)	SERT + NET inhibition	↑ 5-HT, NE	Serotonergic load
LSD (200 μg)	5-HT2A agonism	↑ 5-HT2A	**Triple serotonergic**
Ketamine (245 mg)	NMDA antagonism	Dissociation	Respiratory risk + 5-HT
Prazepam (40 mg)	GABA-A positive mod.	CNS depression	+ ketamine = resp. risk
Cyamemazine (37.5 mg)	D2/5-HT2 antagonism	Sedation	Additive CNS depression
THC (3 chambers)	CB1 agonism	Sedation/dissociation	Additive with ketamine
Nicotine (22 puffs)	nAChR agonism	Sympathetic activation	Cardiovascular stress

Critical risks identified: (1) Triple serotonergic load (sertraline + venlafaxine + LSD) creating serotonin syndrome risk; (2) Triple CNS depression (prazepam + ketamine + cyamemazine) creating respiratory depression risk; (3) Dissociative potentiation (ketamine + THC + LSD);

(4) 4× the prescribed prazepam daily dose in 3 hours. The Garmin stress score remained at 79 throughout this period, while Body Battery was at 5/100 (minimum), suggesting severe physiological depletion that the subject continued to override pharmacologically.

This night exemplifies the gap between the platform's monitoring capability and clinical action. The interaction engine would flag multiple critical/high-severity pairs, but no automated intervention occurred. In a fully deployed system, this pattern should trigger an immediate clinical alert.

A.8 Temporal Pattern Analysis

Of the 181 substance events, **72%** (130 events) occurred between 22:00 and 10:00, indicating predominantly nocturnal consumption. This pattern is clinically significant: nocturnal substance use (a) disrupts sleep architecture, (b) occurs when clinical supervision is unavailable, (c) may indicate insomnia-driven self-medication, and (d) correlates with the sustained low Body Battery readings (5/100 for 80% of the observation period).

A.9 Body Pillar: Movement and Physical Practices

A.9.1 Running

Frequency: ∼9 km/week (P57 in 30–34M cohort). **Mechanism:** Endorphin release, BDNF upregulation, cardiovascular conditioning. **Biometric signature:** HR max tracked via Garmin, post-run HRV recovery measured, Body Battery drain/recharge quantified.

Running serves as the primary endogenous tool for mood regulation. The platform records correlations between running days and next-day biometric readings (HRV and stress baselines); whether these reflect a causal dose–response relationship has not been assessed.

A.9.2 Dance as Awareness Space Vector

Mechanism: Rhythmic full-body movement recruits motor cortex, cerebellum, and basal ganglia simultaneously; engages proprioceptive feedback loops; combines cardiovascular exertion with coordination, musicality, and foreign-language cognitive load. **Biometric signature:** HR elevated to training zone, stress reduction post-session, acute HRV improvement.

Video analysis: Şımarık (Tarkan, Turkish pop, 4:03). Frame-by-frame analysis (20 frames at 12s intervals) of a recorded session reveals the A&B thesis in its most literal form: the body as a physical vector transmitting an awareness space proposition.

Dominant movement patterns:

- **Squat pattern (60% of frames):** Persistent wide stance with center of gravity below hip level. Quadriceps, adductors, and glutes under sustained isometric and eccentric load throughout— equivalent to a 4-minute bodyweight squat endurance protocol.
- **Dynamic lunge pattern (25%):** Forward, reverse, and lateral lunges in continuous motion. Peak amplitude at 2:12 approaches a near-split lateral position under active muscular control—not passive flexibility, but loaded mobility.
- **Hip rotation and torsion (15%):** Oblique engagement through torso-pelvis dissociation. The "şımarık" (spoiled/playful) character of the song is expressed through hip-dominant movements—the body communicates the song's emotional register directly.

Cognitive multitasking (Mind pillar): The session is performed to Turkish-language music. The subject simultaneously: (1) decodes foreign phonemes (Turkish vowel harmony, the soft "ğ", agglutinative morphology); (2) maintains choreographic sequence memory; (3) synchronizes movement to syncopated rhythm; (4) projects emotional expression outward; (5) manages cardiovascular effort. This is *multitasking of consciousness*—not productivity multitasking, but simultaneous engagement of linguistic, motor, musical, proprioceptive, and expressive awareness channels.

Proprioceptive consciousness: Multiple frames capture the subject touching abdomen or thigh mid-movement—reconnection gestures that maintain body awareness during high-intensity output. Barefoot practice on hardwood floor provides unmediated ground contact, engaging plantar proprioceptors that are suppressed by footwear.

Biometric integration: The Garmin wearable (visible in frames) records the session in real time. The same device that tracks substance tapering safety captures this dance session's cardiovascular signature— illustrating that body optimization and consciousness practice produce data on the same biometric substrate.

Energy system: The choreography exhibits natural interval structure—high-intensity phases (deep squats, explosive transitions) alternating with relative recovery (higher stance, reduced range of motion)—producing a cardiovascular training effect without deliberate periodization. This mirrors the organic intensity modulation observed in traditional folk dance forms.

Expand Yourself (Naâman, French reggae, 4:01). The second session uses a French reggae track whose theme—personal expansion and mutual connection ("*Nous nous regardons, nous nous rayonnons*")—invites radically different movement vocabulary. The choreography is *jump-dominant*: explosive vertical movements, wide-legged landings, and dynamic weight transfers produce moderate-to-high cardiovascular demand (contrast: Şımarık's squat-dominant, moderate demand). Primary muscle recruitment shifts toward calves and hip flexors alongside quadriceps and glutes. The reggae off-beat requires internalization of syncopated rhythm patterns—a different cognitive channel than Turkish agglutinative phonology, but equally demanding. Linguistically, the subject processes French lyrics containing neologisms ("Expandiose") and reflexive constructions ("nous nous regardons") while maintaining jump timing, spatial awareness, and musical phrase tracking. The session's *message*: joyful embodiment of expansion, communicated through arms extended outward, grounded positions connecting to roots, and fluid transitions expressing musical flow.

Soul Plan (Naâman, French ambient reggae, 3:53). The third session inverts the energetic profile. "Soul Plan" is ambient, meditative—nearly all transcribed audio is tagged "musique d'ambiance" by Whisper, with a single spoken phrase: "*Ressentis-vous ?*" (Do you feel it?). The choreography responds with *stability-dominant* movement: wide stances, slight knee bends, controlled transitions. Cardiovascular demand is low; the primary training effect is balance and lower-body isometric strength. The consciousness state cultivated is explicitly meditative—the repetitive ambient music creates a mantra-like quality, and the body maintains grounded positions that resemble standing meditation postures. Cognitive load is minimal (French vocabulary difficulty: easy; memorization load: low), freeing attentional resources for *proprioceptive awareness*—the body becomes its own awareness object.

Comparative A&B profile. Table 14 summarizes the three sessions across A&B dimensions. Together they illustrate, in this single case, that dance, selected with intention, constitutes a *subjective self-assessment of a training protocol*: Şımarık maximizes the Mind pillar (linguistic complexity, multitasking), Expand Yourself maximizes the Body pillar (cardiovascular demand, explosive power), and Soul Plan maximizes the Awareness pillar (meditative state, proprioceptive consciousness). All three sessions score 7/10 on the Strong × Smart index, but via different pathways—consistent with the hypothesis that the body is a vector of awareness space with multiple valid projections.

Table 14: Comparative A&B profile of three dance sessions (same subject, same week).

Dimension	**Şımarık (Tarkan)**	**Expand Yourself (Naâman)**	**Soul Plan (Naâman)**
Genre / language	Turkish pop	French reggae	French ambient reggae
Duration	4:03	4:01	3:53
Dominant pattern	Squat 60%	Jump-dominant	Stability-dominant
Cardio demand	Moderate	Moderate–High	Low
Primary muscles	Quads, glutes, core	Quads, glutes, calves, core	Quads, hamstrings, glutes
Flexibility focus	Hip, thoracic	Hip, ankle, spine	Hip, ankle
Linguistic load	High (Turkish)	Moderate (French + neologisms)	Low (French, repetitive)
Consciousness state	Multitasking (5 channels)	Rhythmic entrainment	Meditative / proprioceptive
A&B pillar emphasis	**Mind**	**Body**	**Awareness**
Strong × Smart	7/10	7/10	7/10

This three-session corpus—captured within a single week—illustrates how the A&B practitioner can *program* different consciousness–body configurations through deliberate song selection. The biometric substrate (Garmin wearable, visible in all sessions) records each configuration's physiological signature, enabling data-driven refinement of the movement practice over time.

A.9.3 Yin-Yang Yoga

Mechanism: Yin (parasympathetic activation, vagal tone, flexibility) + Yang (sympathetic engagement, strength, energy). **Biometric signature:** Real-time HRV monitored during practice; stress levels tracked pre/post.

The yin-yang pairing is deliberate: yin yoga targets the autonomic imbalance identified in the WAIS-IV assessment (chronic sentinel mode = sympathetic dominance), while yang yoga builds the physical resilience that supports tapering. The platform quantifies each session's autonomic impact.

A.9.4 Breathwork and Meditation

Mechanism: Direct ANS regulation via controlled breathing; default mode network modulation via meditation. **Biometric signature:** HRV biofeedback during practice, consistent with Garland et al. [2014].

Mind Protocol includes 30+ guided practice routines, each with biometric tracking. The PREVENT trial evidence [Kuyken et al., 2015a] for MBCT as a tapering support informed these features. The platform includes guided practices inspired by mindfulness-based approaches, though fidelity to the structured MBCT protocol used in the PREVENT trial has not been assessed.

A.10 Mind Pillar: Cognitive Enhancement Stack

Table 15: Active supplement stack (started January 31, 2026).

Supplement	Dose	Mechanism	Target
Lion's Mane	420–1260 mg	NGF/BDNF stimulation	Cognitive recovery
Griffonia (5-HTP)	900 mg	Serotonin precursor	Mood support
Saffron extract	1000 mg	Monoamine modulation	Antidepressant adjunct
Magnesium	Standard	NMDA modulation, GABA	Sleep, stress, recovery
Omega-3 (PiLeJe)	Standard	Anti-inflammatory	Neuroprotection
Dynabiane probiotic	Standard	Gut-brain axis	Mood, inflammation

The supplement stack targets the neurobiological substrates that will support eventual SSRI/SNRI tapering: serotonin precursors (griffonia), BDNF upregulation (Lion's Mane), and NMDA modulation (magnesium). The 2–4 week onset of Lion's Mane effects is tracked via cognitive task performance and biometric trends.

A.11 Data Visualizations

A.11.1 Substance Use Timeline

Figure 3 displays all 181 logged substance events across the 5-day observation period, revealing consumption patterns, nocturnal clustering, and escalation trajectories that are not apparent from tabular data alone.

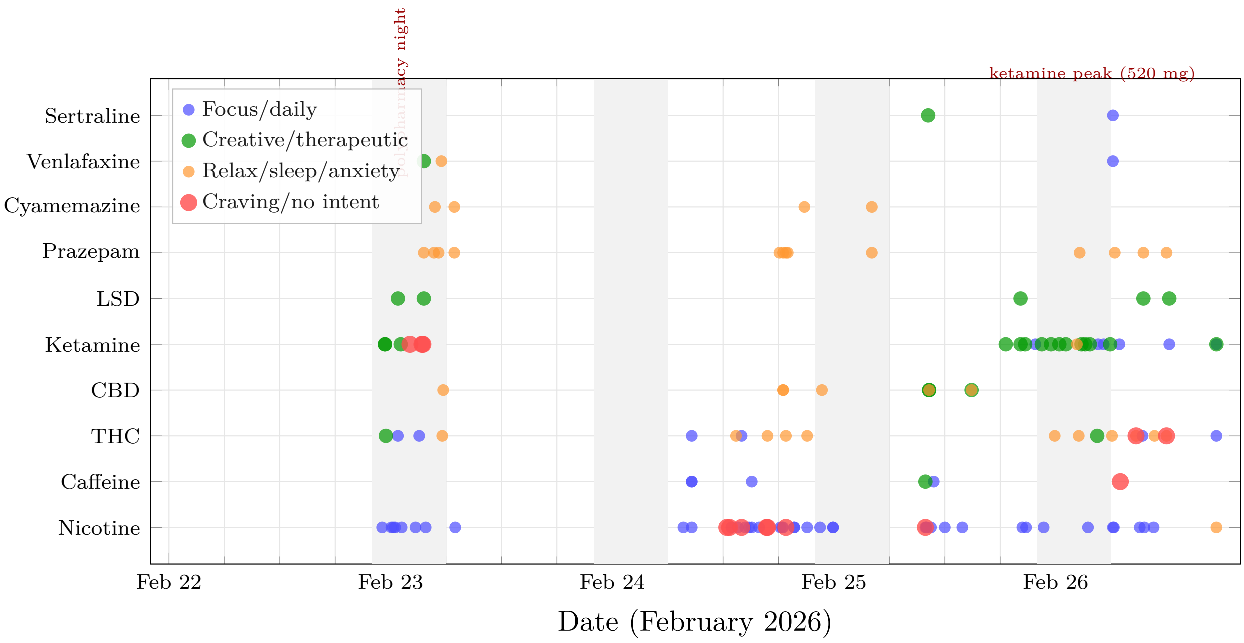

Figure 3: Substance use timeline (Feb 22–26, 2026). Each dot represents one logged administration. Gray bands = nighttime (22:00–06:00). Note the nocturnal clustering and the escalation of ketamine events (y=4) from Feb 22 to Feb 26. Red dots indicate "craving" or absent intent—a dependency signal.

A.11.2 Stress Trajectory with Substance Events

Figure 4 overlays the Garmin stress readings captured at each substance log event, revealing the relationship between consumption intensity and physiological stress.

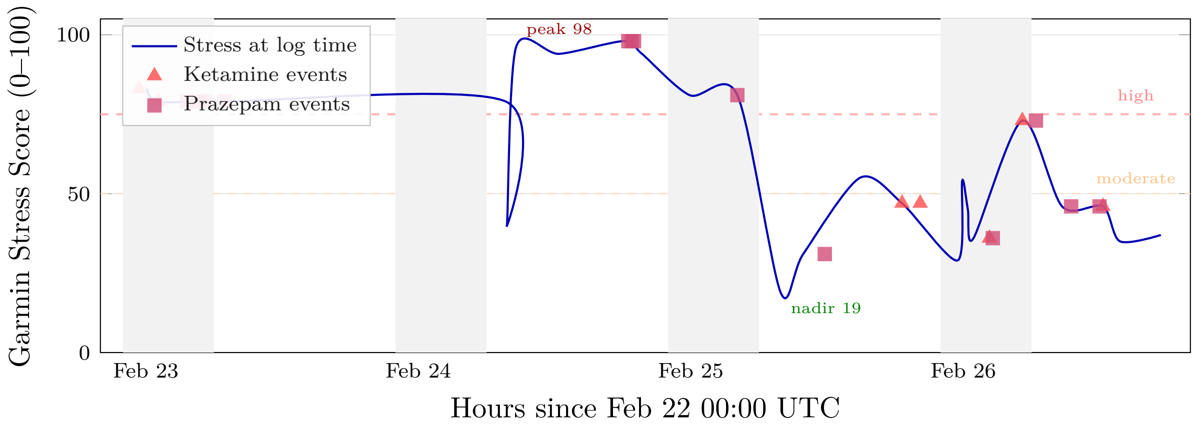

Figure 4: Garmin stress score trajectory overlaid with substance events (Feb 22–26). Stress peaked at 96–98 on Feb 24 (sustained for 12+ hours despite heavy nicotine + THC use), dropped to 19 during a sleep period, then oscillated 29–73 during the Feb 25–26 ketamine escalation. The stress score did not decrease with increasing ketamine consumption, suggesting tolerance or ineffective self-medication.

A.11.3 Dependency Score: Worked Example (Ketamine)

Figure 5 illustrates the dependency scoring algorithm applied to the subject's actual ketamine data.

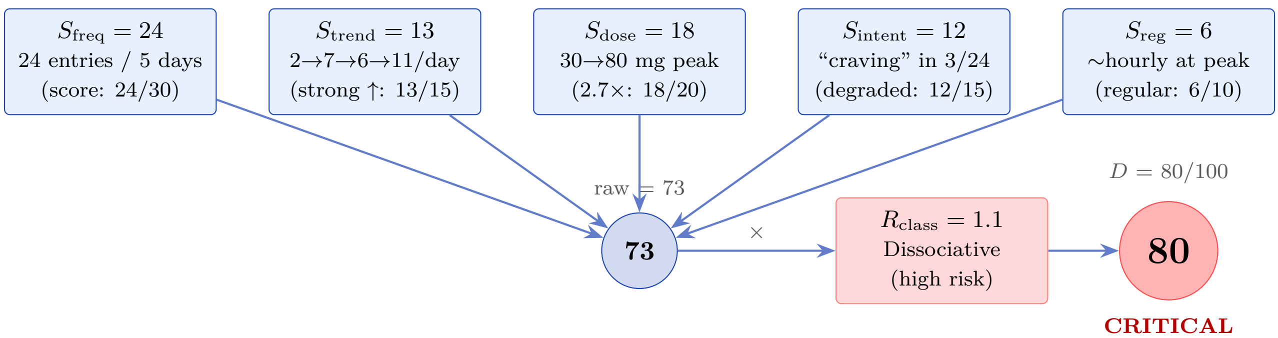

Figure 5: Dependency score computation for ketamine using actual logged data. Raw score (73/90) × risk class modifier (1.1) = 80/100 (critical). Each component is derived from the 24 timestamped entries in Table 10.

A.12 Biometric Baseline (February 2026)

Table 16: Biometric baseline from Garmin Fēnix 8 47mm AMOLED.

Metric	Value	Interpretation
Resting HR	52 bpm	Athletic range
HRV (RMSSD)	44 ms	Low-normal for age
Stress	14/100	Very low (recovery)
Body Battery	93/100	Excellent reserves
ANS Mode	Recovery	Parasympathetic dominant
Sleep duration	8.5 hr (P74)	Above median
Daily steps	6,500 (P32)	Below median
Floors climbed	14/day (P75)	Above median

The biometric profile reveals a paradox observed in this particular subject: excellent physiological recovery capacity (resting HR 52, Body Battery 93) coexisting with chronic psychological hypervigilance. The HRV of 44 ms is low-normal for a 34-year-old male, likely reflecting the impact of the dual SSRI/SNRI regimen and ongoing substance use on autonomic flexibility. This baseline would serve as the reference point for future tapering decisions. In the proposed protocol (Appendix B), a dose reduction that drops HRV below personal baseline or elevates stress above personal threshold would trigger a taper hold—but this automated feedback loop is not yet active. Currently, biometric data is collected and displayed but not used to drive tapering decisions.

A.13 Tapering Roadmap

The following is a *proposed* long-term tapering sequence, informed by the evidence reviewed in this paper. **None of these phases have been initiated at the time of writing.** The subject is currently in a pre-tapering phase (monitoring and logging only):

1. **Phase 1 (Current):** Stabilize ketamine dependency via monitored tapering. Track dose–HRV correlation to establish minimum effective dose. Target: transition from dependency to intentional tool use (PRN, not daily).
2. **Phase 2:** Nicotine tapering guided by stress biometrics and craving intent tracking. Replace with breathwork when possible (same anxiolytic effect, superior HRV profile).
3. **Phase 3:** Evaluate THC dependency vs. tool use status via biometric drift analysis. If tool: maintain as PRN. If dependency: initiate gradual taper.
4. **Phase 4:** Sertraline hyperbolic taper (per Horowitz protocol), supported by established supplement stack, MBCT practices, and continuous HRV monitoring. Minimum 6–12 months.
5. **Phase 5:** Venlafaxine taper (last due to SNRI withdrawal severity). Supported by all preceding tools and practices.
6. **Phase 6:** PRN-only medication profile. All substances available as tools, none required as maintenance. Zero dependencies. Verified by: no dose escalation, no intent degradation, no compulsive timing, stable HRV baseline.

A.14 Proposed Outcome Metrics

These metrics are proposed for longitudinal self-tracking in this individual case. Their validity as clinical endpoints has not been established.

Progress toward zero-dependency is quantified through:

- Number of substances showing dependency signals (target: 0)

- Resting HRV trajectory (target: progressive improvement toward age-appropriate norms)
- Working memory re-assessment (WAIS-IV IMT re-test at 6–8 weeks post-stabilization; target: ≥100)
- Intent specificity ratio: proportion of substance logs with specific functional intents vs. vague/craving intents (target: >90%)
- PRN medication frequency: trending down over months
- Body Battery and sleep quality: maintained or improved during each tapering phase

B Technical Appendix: Mind Protocol Implementation Parameters

This appendix documents the exact parameters extracted from Mind Protocol's source code (`server.py`), enabling independent replication and audit. All values are from the production codebase as of February 2026.

B.1 Dependency Scoring Algorithm

The composite dependency score $D \in [0, 100]$ is computed as:

$$D = \min\left(100,\ \left(\underbrace{S_{\text{freq}}}_{\text{frequency}} + \underbrace{S_{\text{trend}}}_{\text{freq. trend}} + \underbrace{S_{\text{dose}}}_{\text{dose esc.}} + \underbrace{S_{\text{intent}}}_{\text{intent shift}} + \underbrace{S_{\text{reg}}}_{\text{regularity}}\right) \times R_{\text{class}}\right) \tag{1}$$

where:

Table 17: Dependency scoring components and ranges.

Component	**Range**	**Description**
S_{freq}	0–30	Raw frequency of substance use (logs/period)
S_{trend}	0–15	Trend direction of use frequency (increasing = higher)
S_{dose}	0–20	Dose escalation over time (dose at end vs. start)
S_{intent}	0–15	Intent degradation (specific → craving/habit)
S_{reg}	0–10	Regularity of timing (fixed intervals = higher risk)
R_{class}	0.3–1.3	Risk class modifier (supplements: 0.3; opioids: 1.3)

Maximum raw score = 90 × 1.3 = 117, clamped to 100. Risk thresholds: <20 = low, 20–40 = moderate, 40–70 = high, >70 = critical.

B.2 Pharmacokinetic Duration Model

The `PK_DURATION_MIN` map specifies minimum pharmacokinetic durations (in minutes) used to enforce safe intervals between dose changes:

Table 18: Pharmacokinetic duration parameters (selected substances).

Substance	Min	Substance	Min	Substance	Min
Nicotine	30	Caffeine	300	LSD	720
Ketamine	120	THC	240	Psilocybin	360
Alcohol	180	CBD	360	MDMA	360
Cocaine	60	Prazepam	720	Sertraline	1,440
Amphetamine	480	Venlafaxine	720	Cyamemazine	720

B.3 Drug Interaction Matrix

The `SUBSTANCE_INTERACTIONS` matrix encodes 47 pairwise drug–drug interactions with severity levels. Selected examples:

Table 19: Selected entries from the 47-interaction matrix.

Substance A	Substance B	Severity	Mechanism
Sertraline	MDMA	Critical	Serotonin syndrome risk
Sertraline	LSD	Moderate	Serotonergic interaction
SSRI	MAO inhibitor	Critical	Hypertensive crisis
Prazepam	Alcohol	High	CNS depression potentiation
Venlafaxine	Ketamine	Moderate	Serotonergic + dissociative
CBD	Sertraline	Moderate	CYP2D6 inhibition
THC	Cyamemazine	Moderate	Additive sedation
Ketamine	Alcohol	High	CNS/respiratory depression

B.4 Tapering Protocol Specifications

The `TAPERING_PROTOCOLS` module defines 21 substance-specific tapering protocols. Each specifies:

Table 20: Tapering protocol parameters by risk class.

Risk Class	Step (%)	Interval (d)	Hold trigger	Substances
Critical	5–10	7–14	Any threshold breach	BZDs, barbiturates
High	10–15	14–28	$\geq$2 markers	SSRIs, SNRIs, APs, opioids
Moderate	15–20	7–14	$\geq$2 markers	Gabapentin, pregabalin
Standard	20–25	3–7	$\geq$3 markers	Caffeine, nicotine
Minimal	25–50	1–3	Not monitored	Supplements

B.5 Biometric Validation Thresholds

The `BiometricValidation` module defines per-substance deviation thresholds from personal baselines that trigger taper holds or alerts:

Table 21: Per-substance biometric withdrawal detection thresholds.

Substance	**Stress Δ**	**HRV Δ**	**Sleep Δ**
Sertraline	+10	−8	−10
Venlafaxine	+15	−10	−15
Prazepam	+25	−15	−20
Cyamemazine	+15	−10	−15
Ketamine	+20	−12	−15
THC	+15	−8	−10
Nicotine	+10	−5	−8

Where Δ represents the deviation from the individual's 7-day rolling baseline. Positive stress Δ = increase above baseline; negative HRV/sleep Δ = decrease below baseline. When a threshold is breached, the protocol triggers a hold (pause current taper step) and generates a clinical alert. Two consecutive breach-free intervals are required before resuming the taper.

B.6 Surveillance Symptom Categories

73 withdrawal symptoms are monitored across 8 categories:

1. **Neurological** (12): headache, dizziness, paresthesia, brain zaps, tremor, ...
2. **Gastrointestinal** (8): nausea, vomiting, diarrhea, appetite changes, ...
3. **Cardiovascular** (6): palpitations, tachycardia, blood pressure changes, ...
4. **Psychological** (14): anxiety, irritability, insomnia, vivid dreams, depersonalization, ...
5. **Sensory** (8): visual disturbances, tinnitus, hypersensitivity, ...
6. **Motor** (7): akathisia, myoclonus, gait instability, ...
7. **Cognitive** (9): concentration difficulty, memory impairment, confusion, ...
8. **Autonomic** (9): sweating, temperature dysregulation, dry mouth, ...

Symptoms are tracked via daily patient self-report and correlated with biometric data streams. The structured tracking addresses the finding by Henssler et al. [2024] that structured assessment detects significantly more withdrawal cases than unstructured clinical evaluation.

B.7 Pseudo-Code: Taper Step Decision

The core decision algorithm for each taper step evaluation:

```
function evaluate_taper_step(patient, substance):
    baseline = get_rolling_baseline(patient, days=7)
    current  = get_current_biometrics(patient)
    thresholds = BIOMETRIC_THRESHOLDS[substance]

    stress_delta = current.stress - baseline.stress
    hrv_delta    = current.hrv - baseline.hrv
    sleep_delta  = current.sleep_score - baseline.sleep_score

    breaches = 0
    if stress_delta > thresholds.stress_increase: breaches += 1
    if hrv_delta    < thresholds.hrv_decrease:    breaches += 1
    if sleep_delta  < thresholds.sleep_decrease:  breaches += 1

    protocol = TAPERING_PROTOCOLS[substance]

    if breaches >= protocol.hold_trigger:
        return HOLD  // pause taper, alert clinician
```

```
    elif dependency_score(patient, substance) < 20:
        return COMPLETE  // taper finished
    else:
        next_dose = current_dose * (1 - protocol.step_pct)
        return REDUCE(next_dose, protocol.interval_days)
```